5. Decision Networks

CSC3043S Reasoning in AI: Bayesian Networks

Dr Jan Buys

University of Cape Town

[Based on slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All materials available at http://ai.berkeley.edu.]

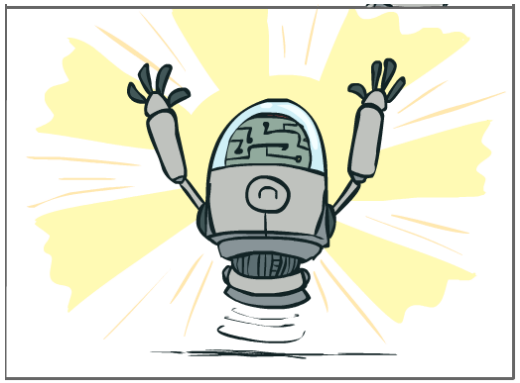


UNIVERSITY OF CAPE TOWN

IYUNIVESITHI YASEKAPA • UNIVERSITEIT VAN KAAPSTAD

What is AI?

Can define the goal of AI to build agents that act rationally



Define the term rational in a specific, technical way:

- **Rational:** maximally achieving pre-defined goals
- Rationality only concerns what decisions are made, not the thought process behind them
- Goals are expressed in terms of the **utility** of outcomes
- Being rational means ***maximizing expected utility***

Rational agents

- Many AI problems can be framed as decision making
- The goal of a rational intelligent agent is to choose actions which **maximize its expected utility, given its knowledge**
- Where do these utilities come from?
- How can we extend Bayesian networks to use them to make decisions which maximizes expected utility?

Rational agents

Utility-based agents:

- A utility function indicates a measure of success or happiness when at a given state
- This can measure the *degree* to which a goal is achieved
- Allows decisions that involve trading off conflicting goals, and choosing between the likelihood of success and importance of a goal
- The agent selects the action with the highest **expected utility**, i.e. the action that will be the most likely to result in a future state that is closest to the desired goal state

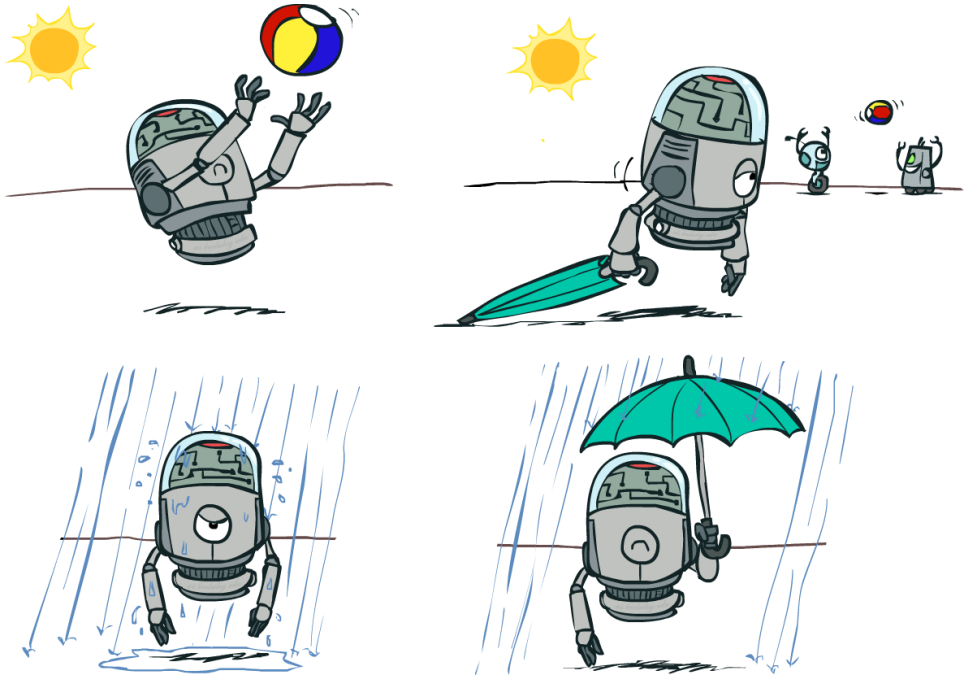
Utilities

- Utilities are functions from outcomes to real numbers that describe an agent's preferences
- Utilities summarize "what the agent wants", i.e., it can be interpreted as giving a preference ranking over outcomes
- Rational preferences are defined in terms of axioms of rationality. This can be shown to be consistent with maximizing expected utility
- The standard approach is to set the utilities manually and let behaviours emerge by optimizing for maximum expected utility

Decision networks

Example: Decide whether to take an umbrella when going outside

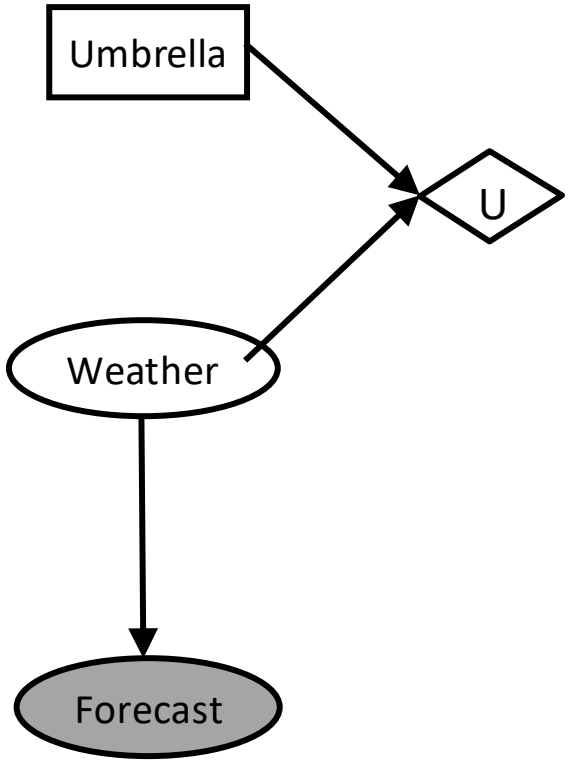
- What is the utility of taking an umbrella when it turns out to rain vs being sunny?



Decision networks

A **decision network** is a Bayesian network with nodes for actions and utility

- Also known as influence diagrams
- We can use decision networks to calculate the expected utility of actions



Decision networks

Node types:



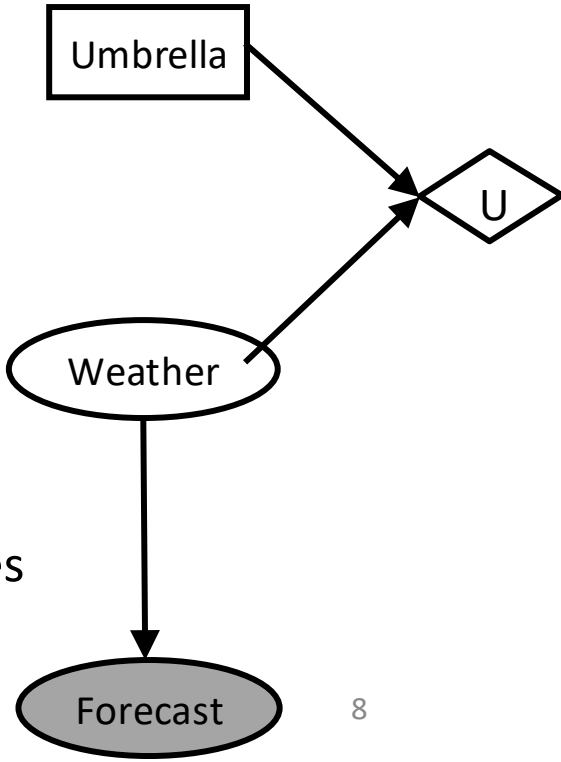
- Chance node (ellipse)
  - Represents observed or unobserved random variables (as in BN)



- Action/decision node (rectangle)
  - Represents decisions (acts as observed evidence)
  - If the action node has chance nodes as a parent: Evidence about the parent node will available when making decision
  - If the node has other action nodes as parents: represents order of decisions



- Utility node (diamond)
  - Depends on action and chance nodes, cannot have children
  - A *utility function* quantifies the usefulness or desirability of the outcomes



Decision networks

Principle of maximum expected utility: *Choose the action which maximizes the expected utility given the evidence*

- The utility function has to be defined by the network designer
- Utility functions can be analogous to money, but there are caveats
- No units, only a scale to distinguish between the utility of the best, neutral, worst outcome
- Best to consider it in relative terms, e.g. 20 is twice as desirable as 10, -20 has the same measure of non-desirability than 20, 0 is usually neutral

Decision networks

Action selection with decision networks:

- Instantiate all the evidence **e**
- For each possible assignment of the action node **a**:
  - Calculate the **posterior distribution** for all the unobserved parents **S** of the utility node, given the evidence: **P(S|a, e)**
    - i.e., perform inference with the parents as query variables
  - Calculate the **expected utility EU(a, e)** for each action
- Choose the action with the maximum expected utility

Decision networks

- Expected utility (EU) of an action a, with evidence variable $E = e$, hidden parent variable $S = s$, and utility function $U(s, a)$:

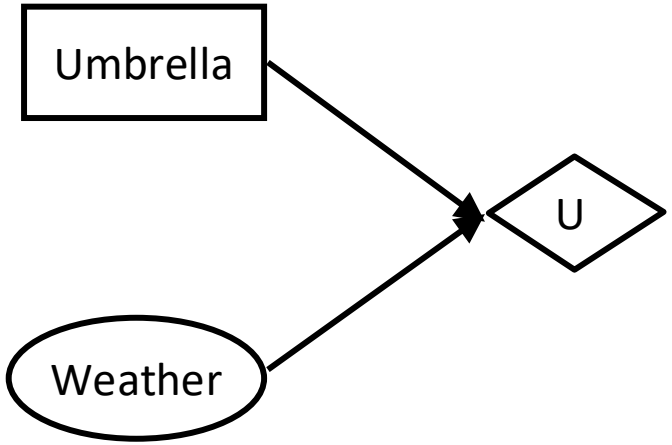
$$\mathrm{EU}(a, e) = \mathrm{E}_{P(S|e)}[U(s, a)] = \sum_{s} P(s|e)U(s, a)$$

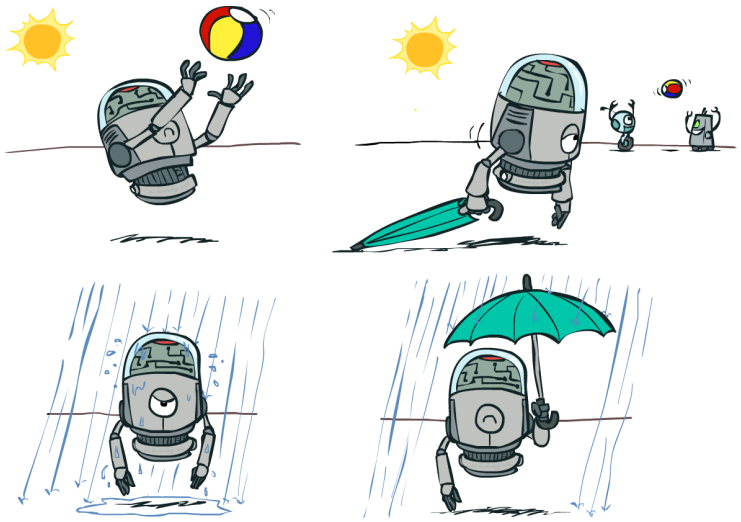
- Maximum expected utility (MEU):

$$\mathrm{MEU}(e) = max_a \mathrm{EU}(a, e)$$

Decision networks

- Example (no evidence):





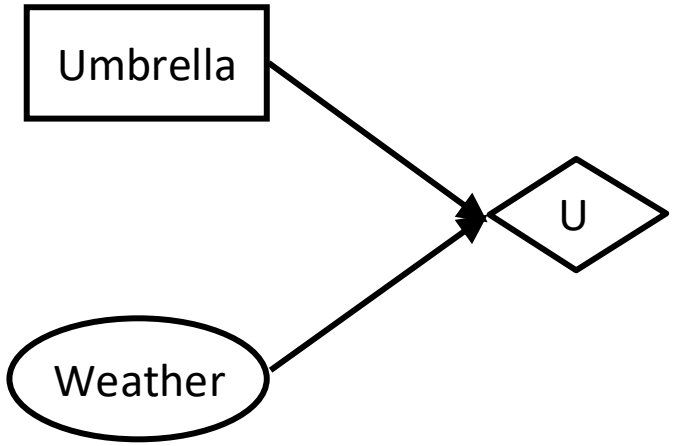
W	P(W)
sun	0.7
rain	0.3

A	W	U(A,W)
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70

Decision networks

- Example (cont):

A	W	U(A,W)
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70



W	P(W)
sun	0.7
rain	0.3

Umbrella = leave

$$\mathrm{EU}(\mathrm{leave}) = \sum_w P(w)U(\mathrm{leave}, w)$$

$$= 0.7 \cdot 100 + 0.3 \cdot 0 = 70$$

Umbrella = take

$$\mathrm{EU}(\mathrm{take}) = \sum_w P(w)U(\mathrm{take}, w)$$

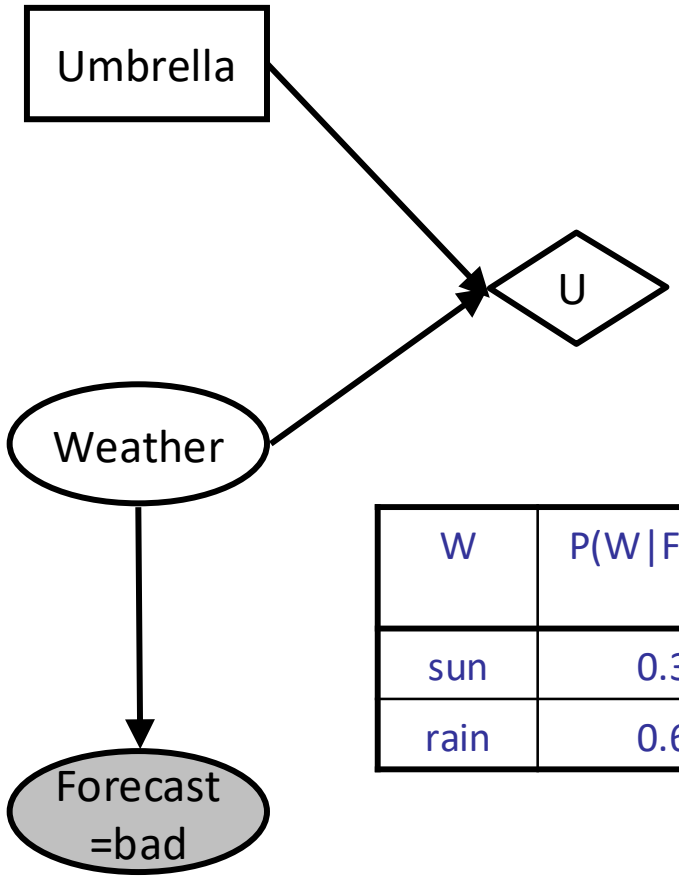
$$= 0.7 \cdot 20 + 0.3 \cdot 70 = 35$$

Optimal decision = leave

$$\mathrm{MEU}(\emptyset) = \max_a \mathrm{EU}(a) = 70$$

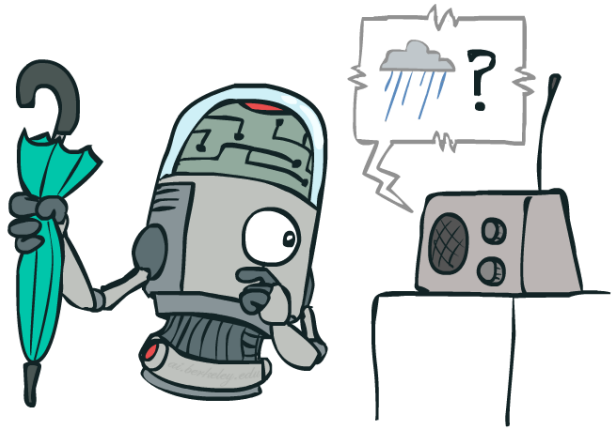
Decision networks

- Example with evidence:



W	P(W|F=bad)
sun	0.34
rain	0.66

A	W	U(A,W)
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70



Decision networks

- Example with evidence (cont):

Umbrella = leave

$$\mathrm{EU}(\text{leave}|\text{bad}) = \sum_w P(w|\text{bad})U(\text{leave}, w)$$

$$= 0.34 \cdot 100 + 0.66 \cdot 0 = 34$$

Umbrella = take

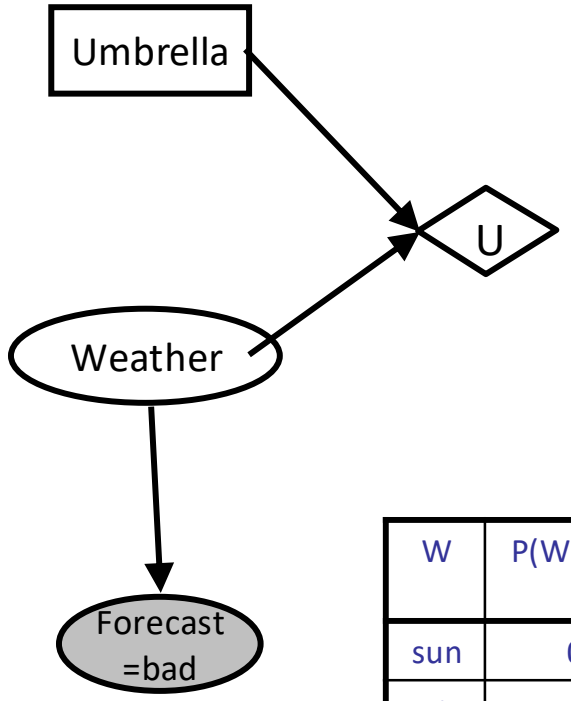
$$\mathrm{EU}(\text{take}|\text{bad}) = \sum_w P(w|\text{bad})U(\text{take}, w)$$

$$= 0.34 \cdot 20 + 0.66 \cdot 70 = 53$$

Optimal decision = take

$$\mathrm{MEU}(F = \text{bad}) = \max_a \mathrm{EU}(a|\text{bad}) = 53$$

A	W	U(A,W)
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70



W	P(W|F=bad)
sun	0.34
rain	0.66

Decision networks

Information links are arcs from chance nodes to decision nodes

- Indicate when information is required/available when a decision is made
- Not a part of the basic evaluation process and have no associated parameters
- With an information link in place, network evaluation can be extended to calculate explicitly what decision should be made, *given* the different values for that chance node

Decision networks

A **decision table** is a table of optimal actions conditional upon the different relevant states of affairs

- To calculate, extend the action selection procedure with another loop
- The conditional decision table is also referred to as a **policy**

Decision networks

Computing the decision table (single action node with information links):

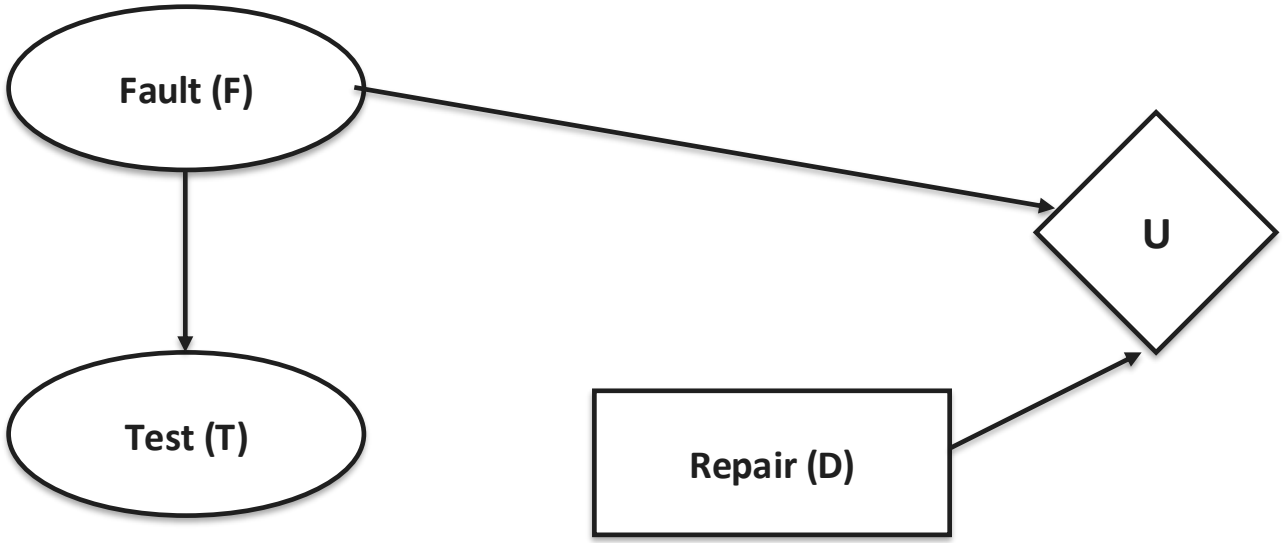
- Instantiate any observed evidence
- For each combination of values of the parents of the decision node:
  - Compute the expected utility of each action as before
  - Record the action that maximizes expected utility

Decision networks

- Example: a machine that may or may not have a fault
  - A factory machine has a hidden fault with probability 0.2.
  - If the machine is faulty and is not repaired, it breaks down mid-production, at a cost of 1000.
  - A repair costs 100, whether or not the fault was actually there.
  - A diagnostic test is available. It is not perfect: it reports positive for 90% of faulty machines, but also for 10% of healthy ones.
- Questions:
  - Should the engineer repair the machine?
  - Should the engineer run the test first? What is the test worth?

Decision networks

The decision network (no test result available yet):



F	P(F)
faulty	0.2
ok	0.8

F	D	U(F, D)
faulty	repair	−100
ok	repair	−100
faulty	don't	−1000
ok	don't	0

F	T	P(T | F)
faulty	pos	0.9
faulty	neg	0.1
ok	pos	0.1
ok	neg	0.9

Decision networks

- The decision with no information
  - The engineer must act knowing only the prior P(F=faulty) = 0.2

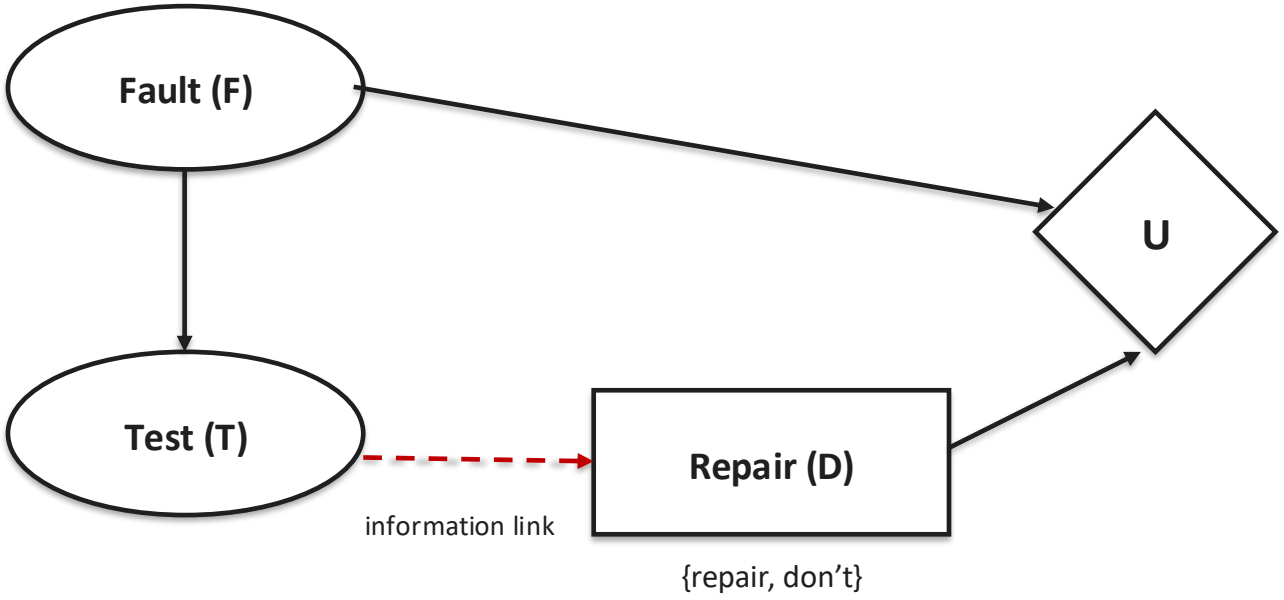
EU(D=repair) = P(faulty)×(−100) + P(ok)×(−100)
= 0.2(−100) + 0.8(−100) = −100

EU(D=don't) = P(faulty)×(−1000) + P(ok)×(0)
= 0.2(−1000) + 0.8(0) = −200

MEU = −100, so with no information the engineer always repairs.

Decision networks

- Now suppose the test is run before deciding: add an **information link** T → D
- The link only says: the value of T will be known at the time the decision is made
- So the answer is no longer a single action, but a decision table (policy) giving the best action for each value of T



Decision networks

- Compute the decision table for each value of T
  - Case T = pos:

P(T=pos) = 0.2(0.9) + 0.8(0.1) = 0.18 + 0.08 = 0.26

P(F=faulty | T=pos) = 0.18 / 0.26 = 0.692

EU(repair | pos) = −100 (repair costs the same either way)
EU(don't | pos) = 0.692(−1000) + 0.308(0) = −692

π(pos) = repair

Decision networks

- Case T = neg:

P(T=neg) = 0.2(0.1) + 0.8(0.9) = 0.02 + 0.72 = 0.74

P(F=faulty | T=neg) = 0.02 / 0.74 = 0.027

EU(repair | neg) = −100
EU(don't | neg) = 0.027(−1000) + 0.973(0) = −27

π(neg) = don't repair

The observation reverses the decision

Decision networks

The decision table (policy) for Repair:

T	P(T)	P(F=faulty | T)	EU(repair)	EU(don't)	π(T)
pos	0.26	0.692	−100	−692	**repair**
neg	0.74	0.027	−100	−27	**don't repair**

Compare with the no-information answer, which was "repair, always".

Decision networks

- Explaining the table
  - Writing p = P(F=faulty | T), *repair* is optimal exactly when

    $-100 > -1000\,p$ i.e. $p > 0.1$

The decision table is just a threshold on the posterior:

T = pos → p = 0.692 (above 0.1) → repair
T = neg → p = 0.027 (below 0.1) → don’t repair

The two observations fall on opposite sides of the threshold

Decision networks

- What is the test worth?
  - Expected value of the MEU of each branch of the decision table over P(T):

MEU with the test = P(pos)×(−100) + P(neg)×(−27)
= 0.26(−100) + 0.74(−27) = −46

MEU without the test = −100

The expected gain from doing the test is: −46 − (−100) = 54

It is worth "paying" up to 54 to run the test before deciding.

Decision networks

- An information link does not always help
  - Same network, but a repair is cheap: U(F, repair) = −5 instead of −100.

Threshold: $-5 > -1000\,p \quad \Rightarrow \quad$ repair whenever $p > 0.005$

T = pos → p = 0.692 → repair
T = neg → p = 0.027 → repair

MEU with the test = −5; MEU without the test = −5

Both observations land on the same side of the threshold, so the policy is constant and the test changes nothing. Information has value only when it can change what you do.

Value of Information

Calculating value of perfect information (VPI):

- Expected value of the MEU when observing variable E':

$$\mathrm{MEU}(E', e) = \mathrm{E}_{P(E'|e)}[\mathrm{MEU}(e', e)] = \sum_{e'} P(e'|e)\mathrm{MEU}(e', e)$$

- Value of perfect information: What is the expected gain in MEU when E' is observed before acting over acting now:

$$\mathrm{VPI}(E'|e) = \mathrm{MEU}(E', e) - \mathrm{MEU}(e)$$

- VPI is non-negative

Value of Information

Example:

- Maximum expected utility (MEU) with no evidence:

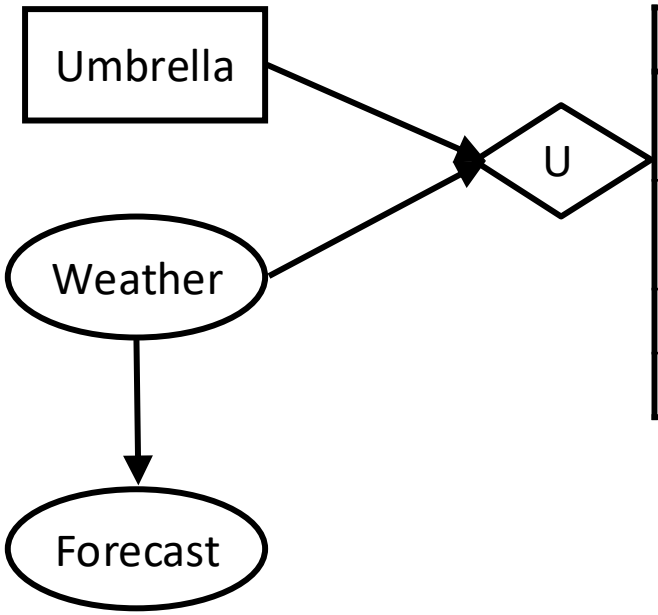
$$\mathrm{MEU}(\emptyset) = \max_a \mathrm{EU}(a) = 70$$

- MEU if the forecast is bad:

$$\mathrm{MEU}(F = \mathrm{bad}) = \max_a \mathrm{EU}(a|\mathrm{bad}) = 53$$

- MEU if the forecast is good:

$$\mathrm{MEU}(F = \mathrm{good}) = \max_a \mathrm{EU}(a|\mathrm{good}) = 95$$



A	W	U
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70

F	P(F)
good	0.59
bad	0.41

Value of Information

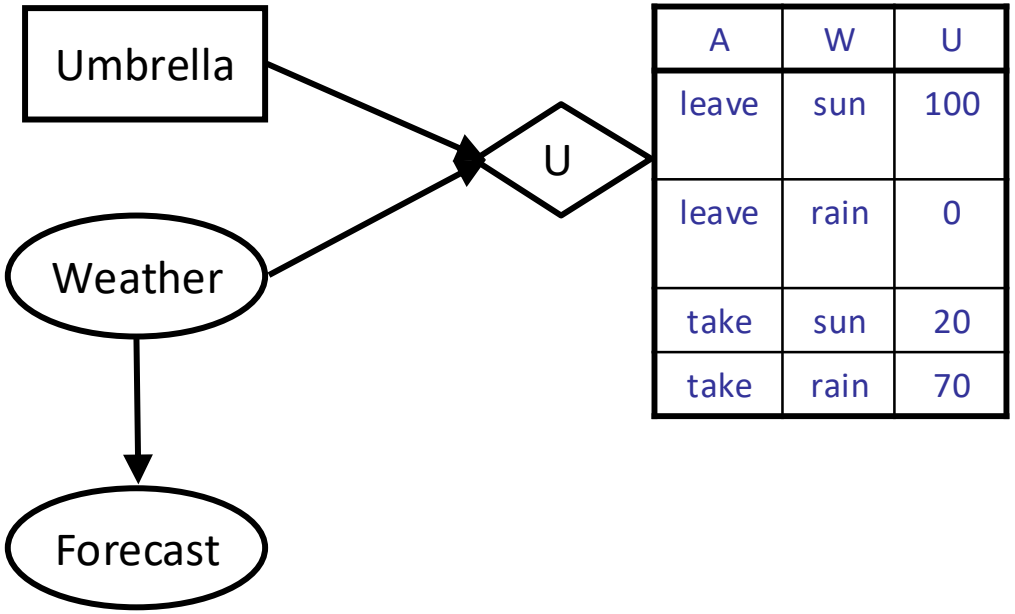
Example (cont):

- Expected MEU with forecast:

$$\mathrm{MEU}(F) = \sum_f P(\mathrm{f})\mathrm{MEU}(f)$$
$$= 0.59 \cdot 95 + 0.41 \cdot 53 = 77.8$$

- Value of information of forecast:

$$\mathrm{VPI}(F) = \mathrm{MEU}(F) - \mathrm{MEU}(\emptyset)$$
$$= 77.8 - 70 = 7.8$$



A	W	U
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70

F	P(F)
good	0.59
bad	0.41